Snap-to-Sell: An Open-Source Pipeline for Turning a Corner-Store Product Photograph into a Retrieval-Grounded E-Commerce Listing

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***Abstract*—Independent retailers increasingly seek to sell inventory through online marketplaces, delivery platforms, and digital storefronts. However, onboarding products remains labor-intensive because each listing requires product identification, pricing, image acquisition, category assignment, and description generation. This study presents Snap-to-Sell, an open-source artificial intelligence pipeline capable of transforming a single smartphone photograph of a packaged retail product into a draft e-commerce listing. The system combines vision-language reasoning, packaging optical character recognition (OCR), retrieval-grounded product search, automated listing generation, catalogue-image verification, and responsible artificial intelligence safeguards. Evaluation was conducted on a hand-labelled dataset of 58 retail products spanning food, beverage, household, personal-care, tobacco, and alcohol categories. Results show improvements over a naive vision-language baseline in product identification, category classification, restricted-item detection, and image-enhancement performance. Snap-to-Sell achieved a top-1 identification accuracy of 86.2%, category accuracy of 79.3%, age-restricted-item recall of 100%, and upgraded 29.3% of evaluated product images. Price estimation emerged as the system's primary limitation due to retrieval ambiguity and multipack matching errors. These findings suggest that retrieval-grounded multimodal systems can meaningfully reduce listing-generation effort while maintaining transparency, traceability, and human oversight.**

***Index Terms*—computer vision, vision-language models, e-commerce, retail analytics, retrieval augmented generation, product recognition, responsible artificial intelligence**

I. INTRODUCTION

A. Background, Context, and Scope

The growth of e-commerce has fundamentally transformed how products are marketed, discovered, and sold. Large retailers typically maintain sophisticated product information management systems and dedicated catalogue management teams. In contrast, independent convenience stores and small retailers frequently rely on manual processes for inventory digitization and online listing creation. Creating a single product listing often requires identifying the product, researching appropriate pricing, selecting images, assigning categories, and writing descriptions. This effort scales poorly across hundreds of products and constitutes a major barrier to digital adoption.

Recent developments in multimodal artificial intelligence have created opportunities to automate many of these tasks. Vision-language models (VLMs) such as CLIP have demonstrated strong zero-shot recognition capabilities by learning aligned visual and textual representations from large-scale image–text datasets [1]. At the same time, retrieval-grounded generation has emerged as an effective strategy for improving factual consistency and reducing hallucination by conditioning language models on external evidence sources [2]. Advances in retail-product recognition datasets such as RP2K and CAPG-GP further demonstrate that fine-grained product identification can be achieved using modern multimodal approaches [**?**], [3].

This project focuses on packaged consumer products commonly found in convenience stores, including food, beverages, household goods, personal-care products, tobacco products, and alcohol. Rather than proposing a novel machine-learning model, the project investigates whether existing AI systems and public data sources can be integrated into a reliable workflow capable of transforming a single product photograph into a publishable e-commerce listing.

B. Problem Statement and Motivation

The primary challenge addressed in this project is that retail product onboarding remains largely manual. Although modern multimodal systems can identify products with reasonable accuracy, practical retail deployment requires substantially more than recognition.

The core question is whether a single product photograph can be transformed into a publishable listing that is sufficiently accurate to support commercial use. Such a system must correctly identify the product, retrieve relevant metadata and price references, generate grounded listing copy, choose appropriate imagery, and provide confidence indicators demonstrating why its recommendations should be trusted.

Several technical challenges complicate this objective. First, many packaged products are visually similar and differ only in flavour, package size, quantity, or regional variation. Product-recognition research has shown that such differences make retail classification significantly more difficult than conventional object recognition [3]. Second, retrieval systems often return multipacks, bundles, or incorrect alternatives, creating substantial pricing errors [4]. Third, language models can generate unsupported claims when not constrained by external evidence [2]. Finally, products such as alcohol and tobacco introduce additional compliance and safety requirements. Consequently, any practical solution must prioritize trustworthiness, transparency, and human oversight rather than automation alone.

C. Research Question and Objectives

The primary research question investigated in this project is:

How accurately can a vision-language and retrieval pipeline identify packaged corner-store products from a single image and generate a publishable, correctly priced listing, and where does it systematically fail?

To answer this question, the project establishes four objectives:

- Build an end-to-end pipeline capable of transforming a product photograph into a structured retail listing.
- Quantify system performance through identification accuracy, pricing error, listing quality, image-upgrade success, and responsible-AI metrics.
- Characterize major failure modes including look-alike products, multipack ambiguity, image quality issues, and restricted-item handling.
- Integrate responsible-AI safeguards capable of detecting age-restricted products, identifying unsupported claims, and routing uncertain predictions to review.

D. Project Contributions

This work makes five primary contributions.

First, it presents a fully integrated and openly documented pipeline capable of transforming a single product photograph into a structured e-commerce listing. Second, it combines vision-language reasoning and OCR-derived packaging information to improve recognition reliability, following evidence that multimodal recognition outperforms image-only approaches in retail settings [5]. Third, it incorporates retrieval-grounded metadata generation using public product databases and shopping-search systems, extending concepts explored in retrieval-augmented listing-generation research [2]. Fourth, it introduces catalogue-image verification through visual similarity matching and perceptual hashing inspired by recent image-retrieval systems [6], [7]. Finally, it demonstrates how responsible-AI safeguards can be integrated into retail workflows through confidence estimation, age-restriction detection, grounding verification, explainability, and human-in-the-loop review.

II. Literature Review

The development of an automated retail listing-generation system sits at the intersection of several active research areas including retail product recognition, vision-language modelling, retrieval-augmented generation, price estimation, product-image retrieval, and responsible artificial intelligence. Although substantial advances have been made within each of these domains individually, relatively little research has evaluated a unified workflow capable of transforming a retail product photograph into a publishable e-commerce listing.

This review examines the major research themes that motivate the Snap-to-Sell architecture and highlights the gap addressed by this work.

A. Retail Product Recognition and Fine-Grained Classification

Retail product recognition presents a specialized form of fine-grained image classification. Unlike traditional object-recognition tasks, retail systems must distinguish among products with highly similar visual appearances, small packaging variations, regional branding differences, and changing inventory assortments.

Large benchmark datasets such as RP2K have demonstrated the complexity of retail-product recognition and established benchmarks for fine-grained classification [3]. Similarly, CAPG-GP highlights the challenges associated with smartphone-captured grocery products under real-world conditions where lighting, occlusion, and camera angle vary substantially [**?**].

Traditional computer-vision solutions relied on supervised classification models trained on large labelled datasets. However, these approaches often struggle when presented with unseen products or packaging variations. Furthermore, maintaining retail-classification systems requires continual dataset updates as packaging evolves and new product variants enter the market.

Consequently, modern retail-recognition research has increasingly shifted toward multimodal approaches that combine visual information with textual signals extracted directly from product packaging [5].

B. Vision-Language Models for Product Understanding

The emergence of vision-language models (VLMs) has significantly changed how image-recognition systems are developed. Models such as CLIP learn aligned visual and textual representations, enabling zero-shot classification and semantic retrieval without task-specific retraining [1].

For retail applications, VLMs provide several advantages. First, they are capable of identifying products through semantic similarity rather than strict category membership. Second, they can generalize more effectively to previously unseen brands and packaging designs. Third, they provide a flexible foundation for generating structured metadata and descriptions directly from images.

These capabilities are particularly important in convenience-store environments where large labelled training datasets may not exist. However, image-only reasoning remains insufficient for reliable product identification because many packaged goods differ primarily through textual details printed on their labels. As a result, recent research has explored the integration of vision-language reasoning with OCR-derived textual features [5].

C. OCR-Augmented Product Recognition

Optical Character Recognition (OCR) has become a critical component of retail recognition systems because a significant portion of product identity is encoded through packaging text.

Brand names, flavour information, nutritional claims, package quantities, and product variants often appear as text rather than graphics. Research has shown that combining OCR-derived textual information with visual representations improves retail-product recognition performance compared with image-only systems [5].

The improvement becomes particularly important when products exhibit nearly identical visual designs but differ through modifiers such as "Diet," "Zero Sugar," "Family Size," or flavour labels. OCR therefore provides additional evidence capable of resolving ambiguities that visual features alone may not distinguish.

The recognition module developed in Snap-to-Sell adopts this multimodal approach by combining vision-language reasoning with packaging OCR prior to retrieval and listing generation.

D. Retrieval-Augmented Product Search

Retrieval-Augmented Generation (RAG) has emerged as a widely adopted strategy for improving the factual consistency of large language models. Instead of relying exclusively on information stored within model parameters, retrieval-grounded systems first obtain supporting evidence from external sources and then generate outputs conditioned on retrieved content [2].

The principal advantage of retrieval grounding is that generated content becomes traceable to supporting evidence. Consequently, retrieval-grounded systems reduce hallucination risk while improving transparency and auditability [2].

For retail applications, retrieval grounding enables product titles, descriptions, prices, and images to be linked to observed evidence rather than generated entirely from model memory. This capability becomes particularly important in commercial environments where inaccurate product information may influence purchasing decisions.

E. Price Estimation and Product Matching

Estimating retail-product prices remains a difficult problem because prices vary across stores, geographic locations, promotions, and package configurations.

Prior research generally follows one of two dominant strategies. The first approach predicts price directly from images or product descriptions using regression methods [**?**]. While such systems can generate plausible estimates, they often provide limited transparency regarding the basis of their predictions.

The second approach utilizes retrieval-grounded pricing. Instead of predicting prices directly, systems retrieve similar products and derive price recommendations from observed market values [4]. This approach improves explainability because pricing decisions can be traced back to specific comparable products.

However, retrieval-based pricing introduces additional challenges. Multipacks, promotional bundles, wholesale cases, and visually similar products frequently appear in search results. As a consequence, retrieval quality becomes the dominant factor controlling pricing performance [4].

These limitations are particularly relevant for convenience-store inventories where low-cost products often appear in multiple package sizes and variants.

F. Product Image Retrieval and Enhancement

Product imagery plays a critical role in e-commerce because images often serve as the first point of customer interaction with a listing.

Recent research has explored embedding-based retrieval systems capable of matching visually identical products across different photographs. Systems such as SIR and e-CLIP demonstrate that similarity-learning approaches can identify semantically equivalent products despite differences in background, viewpoint, or lighting conditions [6], [7].

These capabilities are especially useful when replacing low-quality shelf photographs with higher-quality catalogue images. However, image replacement introduces the risk of incorrect substitutions when visually similar products are mistaken for identical products.

Consequently, recent image-retrieval research increasingly combines similarity matching with verification mechanisms designed to confirm same-SKU equivalence before replacement is permitted [6], [7]. The Snap-to-Sell image-upgrade module follows this principle through the combined use of embedding similarity and perceptual hashing.

G. Responsible Artificial Intelligence in Retail Systems

Responsible Artificial Intelligence (RAI) has become an important consideration in the deployment of commercial AI systems. Retail environments introduce unique risks because inaccurate product information may influence consumer purchasing decisions, violate regulations, or create reputational harm for retailers.

Several themes emerge consistently throughout the literature. First, transparency is required to ensure that users understand how system recommendations are generated. Second, human oversight remains important when confidence is low. Third, confidence-aware routing mechanisms help identify situations in which automated outputs should not be trusted without review. Finally, safety mechanisms are required whenever regulated products are present [2].

These concerns become particularly important in the context of alcohol, tobacco, and age-restricted consumer goods. Consequently, modern commercial systems increasingly combine predictive models with rule-based guardrails, confidence estimation, and human-review workflows.

Table I summarizes the key research streams that contribute to Snap-to-Sell. Existing work typically focuses on a single component of the workflow, such as product recognition, image retrieval, pricing, or listing generation. In contrast, Snap-to-Sell integrates these capabilities into a unified pipeline while incorporating responsible-AI safeguards and human review. This combination of retrieval grounding, image verification, explainability, and publishing-oriented evaluation distinguishes the system from prior research.

H. Research Gap

Although substantial research exists on retail product recognition [**?**], [3], OCR-enhanced classification [5], retrieval-grounded generation [2], price estimation [**?**], [4], and image

TABLE I
REPRESENTATIVE RESEARCH RELATED TO SNAP-TO-SELL

Study	Domain	Primary Contribution	Limitation Relative to Snap-to-Sell
RP2K [3]	Retail Recognition	Large-scale benchmark dataset for fine-grained retail product classification.	Focuses on recognition only; does not address pricing, listing generation, image enhancement, or responsible-AI concerns.
CAPG-GP [?]	Grocery Recognition	Evaluates product recognition under challenging real-world imaging conditions.	Limited to identification tasks and does not consider downstream listing-generation workflows.
OCR-Augmented Recognition [5]	Multimodal Recognition	Combines OCR-derived text with visual features to improve classification accuracy.	Improves recognition performance but does not integrate retrieval, pricing, or publishing workflows.
CLIP [1]	Vision-Language Models	Introduces aligned image-text representations enabling zero-shot recognition and retrieval.	General-purpose model that does not provide retail-specific listing generation or verification capabilities.
ModICT [2]	Listing Generation	Uses multimodal inputs and in-context learning to generate e-commerce product descriptions.	Focuses primarily on description generation and does not evaluate pricing, image verification, or safety controls.
LLP [4]	Price Estimation	Investigates retrieval-grounded pricing using large language models and comparable products.	Targets pricing specifically and does not integrate recognition, image enhancement, or review workflows.
SIR [6]	Image Retrieval	Uses similarity-based retrieval to identify semantically equivalent product images.	Addresses image matching only and does not evaluate listing-generation pipelines.
e-CLIP [7]	Visual Search	Large-scale vision-language representations for e-commerce search and retrieval.	Optimized for retrieval and ranking rather than complete listing generation.
Snap-to-Sell (This Work)	**Integrated Pipeline**	**Combines recognition, OCR, retrieval grounding, price estimation, image verification, responsible-AI safeguards, and human review into a single workflow.**	**Pricing remains vulnerable to multipack ambiguity and retrieval-quality limitations.**

retrieval [6], [7], very little work evaluates a unified pipeline that combines all of these capabilities within a single workflow.

Commercial "snap-to-sell" applications remain largely proprietary, providing limited transparency regarding system architecture, evaluation methodology, retrieval strategies, or responsible-AI safeguards.

The primary contribution of Snap-to-Sell is therefore not the invention of a novel machine-learning model. Instead, the work demonstrates how existing multimodal systems, retrieval services, public product databases, image-verification mechanisms, and responsible-AI controls can be combined into a practical retail-listing pipeline and systematically evaluated.

III. METHODOLOGY

This project follows the Cross-Industry Standard Process for Data Mining (CRISP-DM) framework, adapted for a system assembled primarily from pretrained foundation models, public product databases, and external retrieval services. Unlike traditional machine-learning projects that focus on developing and training predictive models, Snap-to-Sell emphasizes system integration, retrieval grounding, explainability, and evaluation. The objective is not to develop a novel recognition algorithm but rather to determine whether existing multimodal technologies can be combined into a practical retail-listing workflow that is sufficiently accurate, transparent, and safe for real-world use. The use of retrieval-grounded generation is motivated by recent research demonstrating that external evidence improves factual consistency and reduces unsupported claims in e-commerce content generation [2].

The methodology consists of six stages: problem understanding, data understanding, data preparation, modelling, evaluation, and deployment. Each stage contributes to transforming a single retail product photograph into a publishable e-commerce listing.

The six-stage architecture shown in Fig. 3 follows a sequential workflow beginning with image capture and ending with human review and publication. Each stage operates as an independent module connected through structured artifacts, enabling pipeline components to be evaluated and replaced independently.

The artifacts exchanged between modules are illustrated in Fig. 4. The explicit representation of intermediate data improves maintainability, testing, traceability, and explainability, characteristics increasingly recognized as important within responsible-AI systems [2].

A. Problem Understanding

The project investigates whether a single smartphone photograph of a packaged retail product can be automatically converted into a publishable e-commerce listing. Success requires accurate product identification, retrieval of relevant metadata, defensible pricing estimation, grounded listing generation, appropriate imagery, and safeguards capable of preventing unsafe or misleading outputs.

Unlike traditional image-classification tasks, the objective extends beyond recognizing the correct product category. The system must identify the specific SKU, support pricing decisions with evidence, expose uncertainty, and route low-confidence cases to human reviewers. Consequently, the solution is evaluated as a complete workflow rather than as an isolated recognition model.

The primary challenge lies in balancing automation with trust. Although modern vision-language models can identify retail products with increasing accuracy, real-world retail deployment

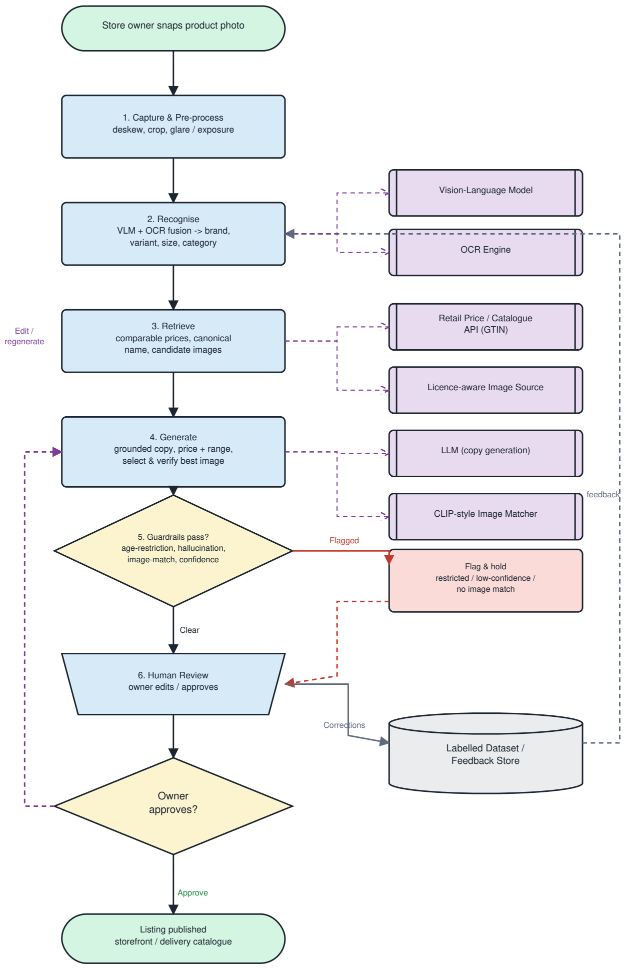

Fig. 1. End-to-end architecture: capture, recognise (VLM + OCR), retrieve (price, canonical name, candidate images), generate, guardrails, human review, and publish, with a feedback loop to a labelled store.

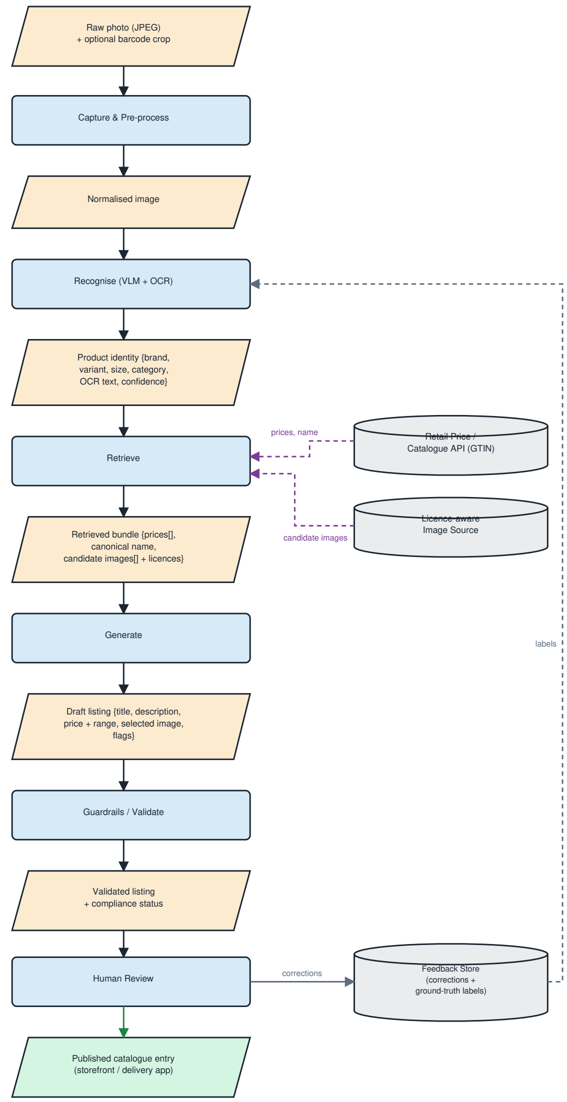

Fig. 2. Data-flow view: the artifacts passed between stages, from raw photo to published catalogue entry, with external datastores feeding retrieval.

requires traceability, transparency, and mechanisms capable of communicating uncertainty [?], [3].

B. Data Understanding

The system operates on product photographs supplied by users and combines them with external information sources. Primary retrieval sources include Open Food Facts, Open Products Facts, Open Prices, and shopping-search systems. These sources provide canonical product names, metadata, candidate product images, and price references used throughout listing generation.

The use of external retrieval sources aligns with retrieval-grounded generation approaches that seek to improve factual accuracy by conditioning outputs on external evidence rather than relying solely on language-model parameters [2].

Exploratory analysis focused on identifying conditions likely to reduce performance. Particular attention was given to visually similar products, multipack configurations, own-brand products, alcohol products, tobacco products, and low-quality photographs. These factors were expected to introduce ambiguity into recognition, retrieval, and pricing stages.

C. Data Preparation

A hand-labelled evaluation dataset consisting of 58 packaged consumer products was constructed. Each product was associated with a verified identity and an observed retail price used as ground truth during evaluation.

The dataset covers six categories:

- Food
- Beverages
- Household Products
- Personal-Care Products
- Tobacco Products
- Alcoholic Beverages

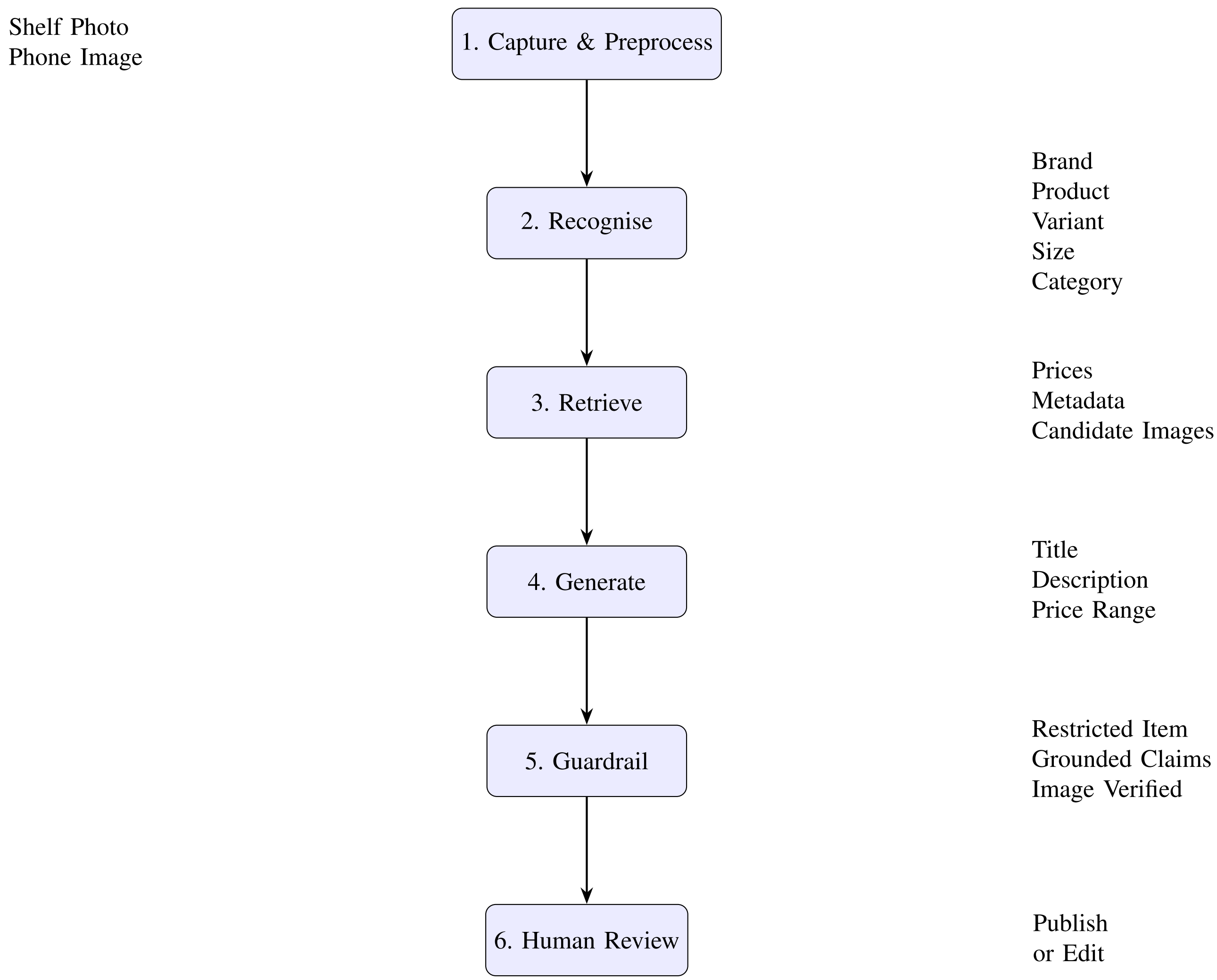

Fig. 3. Snap-to-Sell end-to-end architecture. A product image passes through preprocessing, multimodal recognition, retrieval grounding, listing generation, responsible-AI validation, and human review before publication.

Each item was manually reviewed to verify product identity, category membership, and price information. These labels served as the reference standard for evaluation.

The dataset intentionally includes visually similar products and age-restricted categories to stress-test system behaviour under realistic retail conditions.

D. Modelling Strategy

Several architectural variations were considered during development.

Recognition experiments compared pure vision-language reasoning against multimodal recognition combining OCR-derived text and visual information. This design was motivated by research demonstrating that OCR-enhanced approaches improve retail-product recognition when critical attributes are encoded through packaging text rather than visual appearance [5].

The recognition component additionally draws upon the strong zero-shot recognition capabilities demonstrated by CLIP and related vision-language architectures [1].

Pricing strategies included both direct language-model estimation and retrieval-grounded estimation based on comparable products. Retrieval-based approaches have been shown to improve interpretability because recommendations can be traced to observable comparables and supporting evidence [4].

The final architecture combines:

- Multimodal Recognition
- OCR-Based Information Extraction
- Retrieval-Grounded Generation
- Catalogue-Image Verification
- Confidence Estimation
- Responsible-AI Safeguards
- Human Review

This design was selected because it provides stronger explainability than an end-to-end black-box approach while allowing individual pipeline components to be evaluated independently.

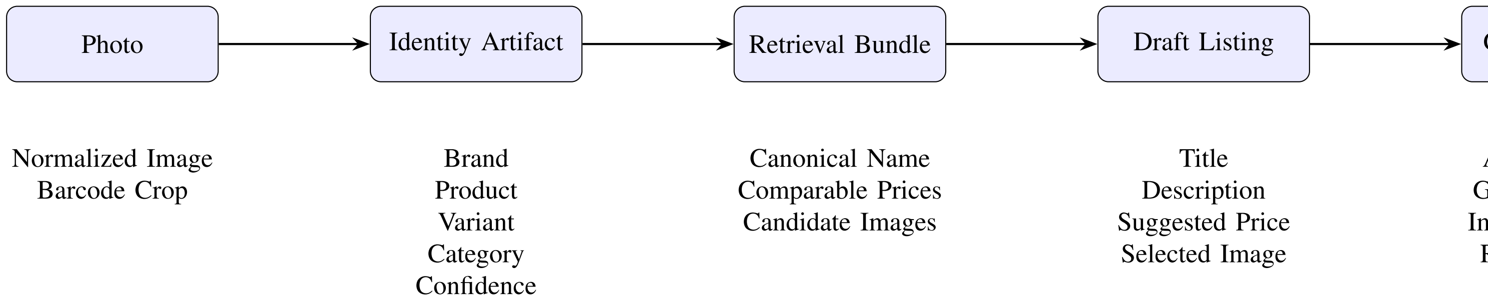

Fig. 4. Data artifacts exchanged between pipeline stages. Each component produces a structured output that becomes the input to the next stage, improving modularity, traceability, and explainability.

E. Evaluation

The completed system is evaluated against a naive baseline consisting of product recognition followed by generic vision-language listing generation without retrieval grounding, image validation, or safety controls.

This evaluation strategy is consistent with recent e-commerce and multimodal-generation research in which improvements are measured relative to simpler generation pipelines rather than through isolated component evaluation [2].

Evaluation focuses on multiple dimensions:

- Product Identification Accuracy
- Category Classification Accuracy
- Price Estimation Error
- Listing Quality
- Image Enhancement Performance
- Responsible-AI Performance
- Processing Efficiency

Failure analysis is treated as a primary outcome rather than a secondary observation. Particular attention is given to retrieval ambiguity, multipack pricing failures, restricted-product handling, and image-substitution risks.

F. Deployment

The deployment stage focuses on academic reproducibility rather than commercial release.

Deliverables include:

- Working Demonstration
- Reproducible Source Code
- HTML Trace Reports
- Evaluation Artifacts
- Presentation Materials
- IEEE-Style Report

This deployment strategy aligns with the CRISP-DM framework while remaining consistent with the educational objectives of the project.

IV. Modelling and Implementation

Snap-to-Sell is implemented as a staged workflow composed of six primary processing modules. The overall architecture follows a human-in-the-loop design philosophy in which automated components generate recommendations while final publishing decisions remain under human control. The workflow integrates multimodal recognition, retrieval grounding, structured listing generation, image verification, responsible-AI safeguards, and explainability mechanisms into a unified end-to-end pipeline.

TABLE II
Technology Stack

Component	Technology
Frontend	Streamlit
Preprocessing	OpenCV
OCR	PaddleOCR
Recognition	GPT-4o-mini
Retrieval	OFF / OPF / Open Prices
Image Matching	CLIP + pHash
Generation	Structured-output LLM
Guardrails	Rule-Based Validation
Reporting	HTML Trace Layer

Rather than relying on a single end-to-end prediction model, Snap-to-Sell decomposes the problem into specialized stages. This design improves transparency, enables modular evaluation, and facilitates future extension of individual components without requiring redesign of the complete workflow.

A. System Architecture

The complete system follows six sequential stages:

1) Capture and Preprocessing
2) Product Recognition
3) Retrieval
4) Listing Generation
5) Responsible-AI Validation
6) Human Review

The architecture is illustrated in Fig. 3. Each stage accepts a structured artifact and produces an output artifact consumed by the next stage. This separation improves maintainability and allows independent evaluation of recognition, retrieval, generation, image verification, and validation modules.

Table II summarizes the principal technologies used throughout the workflow.

B. Capture and Preprocessing

The workflow begins when a retailer captures a photograph using a smartphone camera. The resulting image undergoes several preprocessing operations designed to improve downstream recognition performance.

These operations include:

- Cropping
- Deskewing
- Exposure Normalization
- Glare Reduction
- Optional Barcode Extraction

Preprocessing is implemented using OpenCV and seeks to improve OCR readability while preserving the original characteristics of the captured product image.

Because OCR performance is highly sensitive to image quality, these preprocessing steps improve both text extraction accuracy and downstream product recognition.

C. Recognition Module

The recognition stage combines a hosted vision-language model with packaging OCR. The default configuration utilizes GPT-4o-mini for multimodal reasoning, although the architecture permits alternative hosted models such as Claude or Gemini.

The use of vision-language reasoning is motivated by the strong zero-shot recognition capabilities demonstrated by CLIP and related multimodal systems [1].

The recognition module extracts:

- Brand Name
- Product Name
- Variant
- Package Size
- Product Category
- Confidence Score

OCR complements visual reasoning by extracting textual information that may not be reliably inferred from imagery alone. Prior work has shown that OCR-enhanced approaches outperform image-only systems when important product attributes are encoded through packaging text [5]. This capability is particularly important for products that differ only through flavour labels, package quantity, or limited-edition branding.

The extracted information is serialized into a structured identity object and supplied to the retrieval stage.

D. Retrieval Module

Following recognition, product information is retrieved from multiple external sources. Whenever barcode information is available, the pipeline adopts a barcode-first retrieval strategy because barcode identifiers provide stronger product evidence than image similarity alone.

Candidate records are collected from:

- Open Food Facts
- Open Products Facts
- Open Prices
- Shopping-Search Systems

This retrieval-grounded design follows recent trends in e-commerce content generation that prioritize evidence-supported reasoning over purely generative approaches. By conditioning outputs on retrieved evidence, factual consistency is improved and unsupported claims are reduced [2].

Retrieved candidates are re-ranked according to:

- Brand Consistency
- Category Agreement
- Retailer Relevance
- Currency Compatibility

The retrieval module produces:

- Canonical Product Name
- Comparable Prices
- Candidate Images
- Ambiguity Indicators

These values become the evidence base used by later components.

E. Listing Generation

Generated listings are produced through a structured-output language model.

Unlike generic image-captioning systems, Snap-to-Sell grounds all generated content exclusively on retrieved and extracted evidence.

The generation component follows principles similar to modern retrieval-augmented content-generation systems used within e-commerce applications. Retrieved evidence is supplied directly to the language model so that generated outputs remain grounded in observable product information rather than generated solely from model memory [2].

Generated outputs include:

- Product Title
- Product Description
- Suggested Price
- Price Range
- Listing Metadata

Grounding constraints prevent unsupported specifications, ingredients, or promotional claims from being introduced into generated listings.

This strategy is inspired by recent multimodal product-description generation systems that constrain outputs using structured prompts and external evidence sources [2].

If critical information cannot be verified confidently, generation is deferred and the item is routed to human review.

F. Image Enhancement and Verification

Product imagery plays an important role in e-commerce because images often form a customer's first interaction with a listing.

The image-enhancement module combines two complementary strategies.

When a high-quality catalogue image is available, candidate images are evaluated using:

- CLIP-Style Similarity Scoring
- Perceptual Hashing (pHash)
- Brand Verification
- Category Verification

Replacement only occurs when evidence supports a same-SKU match.

The image-verification strategy follows broader developments in image-retrieval research. Systems such as SIR and e-CLIP demonstrate that similarity-based embeddings can successfully identify semantically equivalent products despite differences in viewpoint, lighting conditions, and image quality [6], [7].

When no suitable catalogue image is available, the retailer's original photograph is enhanced through background-removal techniques while preserving the original product content.

G. Responsible-AI Guardrails

Responsible-AI validation is treated as a first-class system component rather than a post-processing step.

The design follows responsible-AI principles emphasizing transparency, confidence-aware decision making, and human oversight for uncertain recommendations [2].

The guardrail layer evaluates:

- Recognition Confidence
- Retrieval Ambiguity
- Image-Match Confidence
- Unsupported Claims
- Age-Restriction Status

Products identified as tobacco or alcohol automatically receive restriction flags.

Listings containing conflicting evidence, low-confidence identification, or unsupported claims are routed to human review instead of being published automatically.

H. Data Artifacts and Interface Design

A key design feature of Snap-to-Sell is the explicit representation of information exchanged between stages.

Each module produces a structured artifact that becomes the input to the next stage.

The artifacts exchanged throughout the workflow are summarized in Table III.

The explicit representation of intermediate artifacts improves modularity, maintainability, testing, and traceable decision making, which is increasingly recognized as an important requirement of responsible-AI systems [2].

I. Trace Logging and Explainability

A major design objective of Snap-to-Sell is transparency. Rather than treating prediction and generation as opaque processes, the system records evidence produced by every stage of the workflow.

The trace layer captures:

- OCR-Extracted Packaging Text
- Recognition Outputs
- Confidence Scores
- Retrieval Candidates
- Candidate Rankings
- Price References
- Image-Match Scores
- Guardrail Decisions
- Final Recommendations

These artifacts are stored in structured logs and rendered into HTML review reports that can be inspected by human users.

The trace layer serves three purposes.

First, it improves explainability by exposing the evidence supporting recommendations. Users can inspect recognized attributes, retrieval candidates, image-verification scores, and pricing references rather than accepting opaque predictions.

Second, it improves auditability. Because every intermediate decision is recorded, incorrect outputs can be traced back to the specific stage responsible for the failure. This capability supports debugging, quality assurance, and future system refinement.

Third, it enables confidence-aware human review. Products with conflicting retrieval results, uncertain identification outcomes, or unsupported claims are forwarded to reviewers together with the evidence used by the system.

Unlike many end-to-end AI systems, Snap-to-Sell exposes OCR outputs, retrieval evidence, image-verification scores, pricing references, and guardrail decisions. This white-box design improves trust, auditability, and explainability while aligning with responsible-AI recommendations for transparency and traceability in automated decision-support systems [2].

V. Performance Evaluation and Results

A. Evaluation Metrics and Experimental Setup

The objective of the evaluation is to determine whether retrieval grounding, image verification, and responsible-AI controls improve listing quality relative to a simpler baseline workflow.

The baseline system consists of product recognition followed by generic vision-language listing generation without retrieval grounding, catalogue-image verification, or responsible-AI safeguards. Every performance improvement reported by Snap-to-Sell is therefore attributable to the additional retrieval, validation, and review stages introduced by the pipeline.

Evaluation was performed on the hand-labelled dataset consisting of 58 packaged consumer products. The dataset included products from six retail categories:

- Food
- Beverages
- Household Products
- Personal-Care Products
- Tobacco Products
- Alcoholic Beverages

Performance was evaluated across seven dimensions identified during system design:

- Product Identification
- Attribute Extraction
- Price Estimation
- Listing Quality
- Image Enhancement
- Responsible-AI Effectiveness
- Processing Efficiency

TABLE III
ARTIFACTS EXCHANGED BETWEEN PIPELINE STAGES

Stage	Artifact	Contents
Capture	Photo	Normalized image, barcode crop
Recognition	Identity Object	Brand, product, variant, size, category, confidence
Retrieval	Retrieval Bundle	Canonical name, comparable prices, candidate images
Generation	Draft Listing	Title, description, suggested price, selected image
Guardrail	Validation Flags	Age restriction, grounding status, review route
Review	Final Listing	Approved or modified publication-ready listing

TABLE IV
PERFORMANCE ON THE 58-ITEM CORNER-STORE TEST SET

Metric	Baseline	Snap-to-Sell
Top-1 Identification Accuracy	84.5%	**86.2%**
Category Accuracy	77.6%	**79.3%**
Age-Restriction Recall	0%	**100%**
Image Upgrade Rate	0%	**29.3%**
Median Price Error	**22.4%**	43.1%
Mean Price Error	**47.4%**	359.9%

TABLE V
MAJOR FAILURE CATEGORIES

Failure Type	Impact
Multipack Retrieval	Incorrect pricing
Look-Alike Products	Wrong comparables
Regional Variants	Ambiguous matches
Restricted Products	Reduced retrieval coverage
Low Resolution Images	Prevented image replacement

Because several evaluation dimensions remain partially manual, this report focuses primarily on the dimensions for which objective measurements were available at evaluation time.

B. Quantitative Results

The final evaluation was conducted using the deployed Snap-to-Sell pipeline configured with GPT-4o-mini for multimodal recognition and generation, external retrieval services for pricing and metadata retrieval, and CLIP-style image verification for catalogue-image replacement.

Table IV presents the headline results against the naive baseline.

Snap-to-Sell outperformed the baseline on four of the six measured dimensions. The system achieved improvements in top-1 product identification, category classification, restricted-product detection, and catalogue-image enhancement. The strongest gain occurred in responsible-AI performance, where the system correctly identified and flagged every age-restricted product in the dataset.

The image-enhancement module successfully upgraded approximately one-third of all evaluated products through verified catalogue-image replacement. No image replacement occurred within the baseline system because it lacked image-verification capabilities.

Price estimation was the sole dimension in which the baseline outperformed Snap-to-Sell. While retrieval grounding improved explainability, the quality of retrieved comparables significantly affected pricing accuracy. This behaviour is analyzed further in the failure-analysis section.

C. Qualitative Results

In addition to objective metrics, several representative examples illustrate the strengths and limitations of the complete workflow.

1) Successful Listing Generation: A CeraVe Daily Moisturizing Lotion product was successfully recognized, retrieved, and converted into a publishable listing. The system correctly identified the product, retrieved supporting metadata, generated a grounded description, and approved publication without requiring intervention.

2) Age-Restricted Product Handling: An Amarula Cream Liqueur product demonstrated the operation of the responsible-AI layer. The recognition module correctly classified the product as alcohol, the image-upgrade module replaced the original shelf photograph with a verified catalogue image, and the guardrail layer applied an age-restriction flag before publication.

3) Confidence-Based Review Routing: A package of Cream Crackers demonstrated the system's uncertainty-handling capability. Although the product category was recognized successfully, brand verification could not be completed with sufficient confidence. Rather than producing a potentially inaccurate listing, the pipeline routed the item to human review.

These examples illustrate that success is not measured solely by final publication but also by the ability to detect uncertainty and prevent incorrect outputs.

D. Failure Analysis

One of the primary objectives of the project was not merely to evaluate system performance but to characterize where the system fails and why.

Several recurring failure patterns emerged during evaluation.

1) Multipack and Case Pricing Errors: The most significant source of pricing error involved multipack products and wholesale cases returned by retrieval systems.

Shopping-search services frequently returned larger product bundles that were visually similar to the target product. Consequently, retrieved prices sometimes corresponded to 12-

pack, 24-pack, or case quantities rather than individual retail units.

Although price normalization and outlier filtering reduced some errors, multipack ambiguity remained the dominant contributor to poor pricing performance.

2) Look-Alike Product Retrieval: Several pricing failures occurred because retrieval systems returned visually similar but incorrect products.

For example, products sharing packaging colour schemes or category-level similarities occasionally appeared in retrieval results despite belonging to different brands or product families. Although brand-consistency filters eliminated many such cases, some retrieval errors still propagated into the pricing stage.

3) Regional and Unbranded Products: Products with weak branding or region-specific packaging frequently produced ambiguous retrieval results.

When brand verification could not be completed, the confidence layer declined to generate pricing recommendations and routed the product to human review. During evaluation, eight products fell into this category.

4) Age-Restricted Product Retrieval: Alcohol and tobacco products presented a unique challenge because many shopping-search systems intentionally limit indexing of restricted goods.

Although these products often generated low-quality retrieval results, the responsible-AI layer correctly flagged all restricted products in the evaluation dataset. Consequently, safety performance remained strong even when retrieval quality degraded.

5) Catalogue Image Resolution: Another limitation involved image quality.

Shopping-search systems frequently provide small thumbnail images rather than full-resolution catalogue photographs. When candidate replacements failed image-quality thresholds, the system retained the user's original image rather than performing an incorrect or low-quality swap.

E. Discussion and Implications

Several important findings emerge from the evaluation results.

First, retrieval grounding improves trustworthiness but does not guarantee better performance across every dimension. While retrieval enhanced transparency, safety, and image quality, pricing accuracy ultimately depended on the quality of retrieved evidence.

Second, responsible-AI safeguards substantially improved the overall utility of the system. The ability to detect age-restricted products, identify unsupported claims, and route uncertain products to review proved more valuable than maximizing automation alone.

Third, image verification demonstrated that high-quality catalogue-image replacement is feasible when supported by explicit same-SKU validation. The use of CLIP-style similarity scoring and perceptual hashing reduced the risk of incorrect image substitution while improving listing appearance.

F. Interpretability and Explainability

A distinguishing feature of Snap-to-Sell is its emphasis on transparency.

Unlike many end-to-end AI systems that produce opaque predictions, Snap-to-Sell exposes the evidence supporting every recommendation.

Each processed product includes:

- OCR-extracted packaging text
- Recognition outputs
- Ranked retrieval candidates
- Image-matching scores
- Price justification
- Guardrail decisions
- Final publishing recommendation

These artifacts are recorded within a trace log and rendered into an HTML review report, enabling users to inspect every major decision made by the pipeline. This white-box design improves trust, supports debugging, and facilitates human review of uncertain cases.

G. Summary of Findings

Overall, Snap-to-Sell successfully demonstrated that retrieval-grounded, multimodal retail-listing generation is feasible using publicly available data sources and pretrained AI systems.

The system achieved improvements in identification accuracy, category classification, restricted-product detection, and image enhancement while exposing interpretable evidence for every major decision. However, retrieval-grounded pricing remains a significant challenge, particularly for low-cost fast-moving consumer goods where multipack ambiguity and look-alike products dominate retrieval results. These failure modes represent the most important opportunities for future improvement.

H. Project Summary and Key Findings

Snap-to-Sell successfully demonstrated that a single smartphone image can serve as the basis for automated product-listing creation.

The system improved product identification, category classification, image quality, and restricted-item detection compared with a baseline workflow.

I. Conclusions Drawn and Insights Gained

The most important insight is that retrieval grounding improves factual consistency but introduces new challenges associated with retrieval ambiguity.

Successful automation therefore requires both strong retrieval mechanisms and human oversight.

J. Recommendations and Future Work

Future work should focus on:

- Larger retail-product datasets.
- Barcode recognition.
- Package-size verification.
- Marketplace API integration.
- Retail-specific model fine-tuning.
- Multimodal retrieval improvements.

These enhancements may significantly improve pricing reliability and overall listing quality.

VI. Discussion

A. Synthesis of Key Findings

The objective of this project was not simply to determine whether a product could be recognized from a photograph, but rather to evaluate whether a single image could be transformed into a publishable retail listing through a combination of multimodal recognition, retrieval grounding, image verification, and responsible-AI safeguards. The results demonstrate that this objective is achievable for a substantial proportion of packaged consumer products. Snap-to-Sell improved top-1 identification accuracy, category classification, age-restriction detection, and image-upgrade performance relative to the baseline workflow. The strongest improvement occurred in safety-related behaviour, where the system achieved perfect recall for age-restricted items.

The findings also show that retrieval grounding provides benefits beyond raw prediction accuracy. In addition to supplying information used during listing generation, retrieval grounding exposes the evidence supporting generated recommendations. Product metadata, comparable prices, image candidates, and retrieval confidence can all be inspected by human users, making system outputs substantially more transparent than those produced by a purely generative workflow.

At the same time, the evaluation demonstrates that retrieval grounding is not universally beneficial. Although the approach improved trustworthiness and accountability, it introduced new dependencies on retrieval quality. Pricing failures occurred primarily because external search systems returned products that were similar but not identical to the target product. This observation illustrates a broader trade-off between transparency and prediction quality. Transparent retrieval chains expose evidence and uncertainty but remain constrained by the quality of available evidence.

B. Interpretation of System Performance

The strongest-performing components of the system were the multimodal recognition and responsible-AI layers. Identification accuracy exceeded the baseline despite operating on noisy retail photographs captured under real-world conditions. The combination of OCR and vision-language reasoning proved particularly effective because packaging text frequently resolved ambiguities that visual features alone could not.

The image-enhancement module also demonstrated practical value. Nearly one third of all evaluated products received upgraded imagery, improving visual quality without requiring additional user effort. Importantly, image swaps were not performed blindly. Instead, candidate images were verified through a combination of similarity scoring, perceptual hashing, and brand verification before replacement was permitted. This approach reduced the risk of incorrect substitutions while preserving image quality.

The weakest component of the system remained pricing. The evaluation demonstrated that retrieval-grounded pricing could be less accurate than a direct language-model estimate for low-value fast-moving consumer goods. Inspection of the retrieval traces indicates that the majority of pricing errors originated from multipack products, visually similar alternatives, and limited catalogue coverage rather than from generation failures. The pricing stage therefore appears to be constrained primarily by retrieval quality rather than by the language model itself.

C. Practical Implications

Several practical implications emerge from the results.

First, the study suggests that retrieval grounding should be viewed as a trust mechanism rather than purely a performance-improvement mechanism. Grounding provides evidence, traceability, and transparency even when it does not improve every metric. For commercial deployment, these properties may be as valuable as raw predictive accuracy because they enable users to understand and verify system decisions.

Second, responsible-AI safeguards appear essential rather than optional. The ability to identify age-restricted products, prevent unsupported claims, and route uncertain outputs to review significantly improved the reliability of the system. These mechanisms prevented several classes of error from reaching publication despite imperfections in earlier pipeline stages.

Third, explainability provides operational benefits beyond compliance. Because the system records retrieval candidates, OCR outputs, confidence scores, image-verification decisions, and pricing evidence, users can investigate incorrect outputs and understand why specific recommendations were produced. This traceability improves debugging, auditing, and future system development.

D. Limitations and Threats to Validity

The evaluation contains several important limitations.

First, the evaluation dataset consists of only 58 products. While sufficient for a course project and proof-of-concept evaluation, this sample is not large enough to support broad generalization across all retail categories. Additional evaluation on larger and more diverse inventories would improve confidence in the findings.

Second, the system depends heavily on publicly available retrieval sources. Coverage gaps in Open Food Facts, Open Products Facts, and shopping-search systems directly affect downstream performance. Products lacking strong catalogue representation are more likely to experience retrieval failures, pricing ambiguity, or review routing.

Third, image-upgrade evaluation is currently limited by the absence of fully labelled image-swap correctness data. Although swap frequency was measured, future work should quantify false-swap rates directly.

Fourth, listing quality remains partially dependent on human assessment. While objective metrics were available for identification, pricing, and restricted-product detection, more comprehensive user studies would provide additional insight into perceived listing quality and usability.

VII. Conclusion and Future Work

A. Project Summary

This project investigated whether a single smartphone photograph of a packaged consumer product could be transformed into a publishable e-commerce listing through the integration of vision-language models, retrieval grounding, image verification, and responsible-AI safeguards.

The resulting Snap-to-Sell pipeline successfully combines product recognition, metadata retrieval, listing generation, image enhancement, and human review into a unified workflow. Evaluation on 58 retail products demonstrated measurable improvements in product identification, category classification, image-upgrade performance, and restricted-product detection compared with a naive baseline. The responsible-AI layer achieved perfect age-restriction recall and successfully prevented low-confidence products from being published automatically.

B. Conclusions Drawn

The most important finding of this study is that retrieval grounding substantially improves transparency, accountability, and safety but does not guarantee superior performance across all tasks.

Although Snap-to-Sell achieved stronger identification and category performance than the baseline, retrieval-grounded pricing remained vulnerable to multipack products, catalogue inconsistencies, and look-alike retrieval results. These pricing failures represent the most important obstacle to broader deployment.

A second key conclusion is that responsible-AI mechanisms are critical to practical deployment. Rather than maximizing automation, the system prioritizes confidence estimation, evidence presentation, and human review. This design philosophy proved particularly valuable when retrieval quality was insufficient to support reliable automated decisions.

Finally, the study demonstrates that explainability can be integrated into retail AI systems without sacrificing usability. By exposing OCR outputs, retrieval candidates, image-verification evidence, and guardrail decisions, Snap-to-Sell provides users with substantially more visibility into its reasoning process than conventional end-to-end generation systems.

C. Future Work

Several opportunities exist for extending the system.

First, evaluation should be expanded through larger and more diverse retail datasets containing additional brands, product categories, and geographic markets. A larger evaluation set would improve statistical confidence and support more detailed failure analysis.

Second, barcode recognition and package-size verification should be integrated more deeply into the retrieval stage. These enhancements could reduce pricing ambiguity and mitigate the multipack failures observed during evaluation.

Third, future versions of the system should explore learned price-adjustment models capable of correcting retrieval-derived prices. Such models may help compensate for marketplace-specific pricing biases while preserving retrieval transparency.

Fourth, the architecture could be extended to support multi-item shelf detection, allowing multiple products to be extracted and catalogued from a single image. This capability would significantly improve throughput for retail inventory onboarding.

Finally, direct integration with e-commerce platforms and marketplace APIs would allow vetted listings to move seamlessly from review into publication. Combined with continual feedback collection and human corrections, these integrations could provide the foundation for an adaptive retail-listing system that improves over time.

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